# Integrating Prototype Learning With Graph Convolution Network for Effective Active Hyperspectral Image Classification

Chen Ding, *Member, IEEE*, Mengmeng Zheng, Sirui Zheng, Yaoyang Xu, Lei Zhang, *Member, IEEE*, Wei Wei, *Senior Member, IEEE*, and Yanning Zhang, *Senior Member, IEEE*

***Abstract*— In recent years, active learning (AL) methods have provided a feasible approach to alleviate the problem of limited labeled samples in deep learning projects. Existing AL algorithms generally tend to select sample without labeled, whose category is difficult to distinguish. However, the sample in the category center is difficult to determine in AL operations, resulting in inaccurate category measuring and inaccurate sample selection. In addition, hyperspectral images (HSIs) have rich spectral reflective bands with strong correlations, which leads to the phenomenon that the spatial distribution between different categories in HSIs characterizes staggered distribution, which undoubtedly influences the HSI classification effect. In this article, we propose a new AL method (called PLGCN) which combines prototype learning (PL) and graph convolution network (GCN) to solve few-shot HSI classification tasks, and this method can add into existing deep learning-based HSI classification models. It includes two advantages: 1) the prototype of each category is iteratively updated to ensure the optimality of prototype in each sampling stage and 2) the spatial distribution of unlabeled samples is extracted via graph convolution neural network in order to obtain the better features in new space for easier discriminating. Experimental results on three commonly used benchmark HSI datasets demonstrate the effectiveness of the PLGCN in HSI classification tasks with limited labeled samples.**

***Index Terms*— Active learning (AL), graph convolution network (GCN), hyperspectral image (HSI) classification, prototype learning (PL).**

Manuscript received 28 April 2023; revised 18 December 2023; accepted 5 January 2024. Date of publication 10 January 2024; date of current version 23 January 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62372379, Grant 62101454, Grant 62071387, and Grant 61901369; and in part by the National Key Research and Development Program of China under Grant 2022ZD0118700. *(Corresponding author: Yanning Zhang.)*

Chen Ding, Mengmeng Zheng, Sirui Zheng, and Yaoyang Xu are with the School of Computer Science and Technology, Xi'an University of Posts and Telecommunications, Xi'an 710121, China, also with the Shaanxi Key Laboratory of Network Data Analysis and Intelligent Processing, Xi'an 710121, China, and also with the Xi'an Key Laboratory of Big Data and Intelligent Computing, Xi'an 710121, China (e-mail: dingchen@xupt.edu.cn; zmm@stu.xupt.edu.cn; 04204038@stu.xupt.edu.cn; xyy@stu.xupt.edu.cn).

Lei Zhang and Yanning Zhang are with the Shaanxi Provincial Key Laboratory of Speech and Image Information Processing, School of Computer Science, and the National Engineering Laboratory for Integrated Aerospace-Ground-Ocean Big Data Application Technology, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: nwpuzhanglei@nwpu.edu.cn; ynzhang@nwpu.edu.cn).

Wei Wei is with the Shaanxi Provincial Key Laboratory of Speech and Image Information Processing, School of Computer Science, and the National Engineering Laboratory for Integrated Aerospace-Ground-Ocean Big Data Application Technology, Northwestern Polytechnical University, Xi'an 710072, China, and also with the Research and Development Institute, Northwestern Polytechnical University Shenzhen Zhongnan Network Learning Center, Shenzhen 518057, China (e-mail: weiweinwpu@nwpu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3352112

## I. Introduction

PROFITING from the characteristics of hyperspectral images (HSIs) such as high spectral resolution and wide electromagnetic wave range [1]; HSI contains abundant information in spectral and spatial domains [2], [3], [4], and it has been extensively implemented in the fields of land surveying mapping [5], disaster early warning, agricultural planning [6], mineral exploration [7], and so on. HSI classification (HSIC) plays a crucial role in HSI analysis and applications, which dedicates to assigning a specific category for each pixel of the HSI [8], [9], [10]. In the past decades, researchers have strived to solve the HSIC tasks and notched up notable achievements.

Up till the present moment, HSIC methods generally can be divided into nondeep learning-based methods and deep learning-based methods. Initially, researchers utilized nondeep learning-based methods to address HSIC tasks. The majority of nondeep learning-based methods typically comprise two separate parts, handcraft feature extraction, and classifier. Researchers have achieved good performance in both two parts, respectively. Principal component analysis (PCA) [11], independent component analysis (ICA) [12], and linear discriminant analysis (LDA) [13]—these common handcraft feature extraction methods are utilized for extracting the spectral features of HSI. The classifiers mostly use traditional machine learning methods, e.g., support vector machines (SVMs) [14], $k$-nearest neighbor (KNN) [15], and the Gaussian process approach [16]. However, these methods merely rely on isolated pixel spectral information of HSIs [17] without considering the spatial correlation among adjacent pixels, which may lead to unsatisfactory classification results. Therefore, researchers have exploited some methods, which concurrently combine spectral features and spatial features to reinforce the presentation of HSIs and improve classification accuracy [18], [19], [20]. These methods can operate by extracting the spatial features of HSI, and these spatial features usually include extended morphological profile (EMP) [21], local binary pattern (LBP) [22], and simple linear iterative cluster (SLIC) [23]. Although various nondeep learning-based methods have shown their effectiveness unfold before our eyes, they still seem insufficient for further improvement of the classification accuracy of HSIs.

In contrast to nondeep learning-based methods, deep learning-based methods have drawn increasing attention in HSIC applications due to their powerful representation

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:16:59 UTC from IEEE Xplore. Restrictions apply.

ability [24]. Researchers have made a great variety of attempts and have generated many feasible approaches. Chen et al. [25] proposed the stacked autoencoder (SAE) for accurate hyperspectral classification, which is the first employment of deep learning theory in HSIC tasks. In the adjacent year, the researchers from the above team also proposed a deep belief network (DBN) [26] to settle the HSIC task. Subsequently, a lot of methods have been proposed, such as recurrent neural networks (RNNs) [27], different dimensions of convolutional neural networks (CNNs) [28], [29], [30], [31], [32], capsule networks [33], [34], generative adversarial networks (GANs) [35], and transformers [36]. These methods have promoted the improvement of classification accuracy to a certain extent and revealed great potential. Additionally, researchers also introduced different strategies for further enhancing the effect of HSIC, such as residual connection [37], attention mechanism [38], [39], [40], [41], and so on. For instance, Roy et al. [32] proposed a hybrid framework coupling 3-D-CNN and 2-D-CNN, named HybridSN, which can decrease the computational complexity of the model compared with using 3-D-CNN alone. Zhong et al. [37] presented an end-to-end framework named spectral–spatial residual network (SSRN), which utilizes its spectral and spatial residual blocks to obtain extensive discriminative features of HSI. The method in [39] designed a spectral attention module and a spatial attention module for promoting the discriminability of the model and suppressing the impact of useless pixels in the spectrum block.

Although deep learning-based methods have contributed to the boosting of classification accuracy, they are still based on enormous supervision samples for training. When the use of labeled samples is reduced to simulate the application of real situations, the overfitting of network model training will occur and affect the classification accuracy of the model. Active learning (AL) theory can expand the limited labeled sample set by selecting more informative and valuable samples from unlabeled samples. Consequently, AL-based methods are gradually applied to the field of computer vision (CV), especially in the practical application with time-consuming and labor-intensive manual annotations such as HSIC.

According to different sampling strategies, AL methods applied to traditional images can be divided into the following categories, uncertainty-based methods [42], [43], feature space coverage-based methods [44], [45], maximum change based-methods [46], [47], [48], generative adversarial theory-based methods [49], [50], [51], and the method for integrating uncertainty and diversity [52]. The case in an uncertainty-based method such as the method in [42] (i.e., Entropy), which selects the samples to be annotated economically and efficiently by measuring the minimum confidence, marginal sampling, and entropy. Example of the method based on feature space coverage in [44] (i.e., CoreSet), which utilizes the geometric structure of data points to characterize the performance of the selected subset and selects the subset that is expected to produce the best result. The method based on maximum change such as learning loss [46] adds a subsidiary subnetwork for the task model to learn the loss value of the prediction sample and selects the unlabeled sample with the greatest loss. Example of a generative adversarial theory-based method such as variational adversarial AL (VAAL), which utilizes a variational autoencoder (VAE) and a trained adversarial network to distinguish whether samples are labeled. This method selects the sample with the lowest confidence in the discriminator prediction results to expand the limited training set. Recently, graph theory has also been introduced and employed in combination with current AL-based methods. In light of the characteristic of graph convolution network (GCN) that can transfer information between graph nodes and learn the relative relationship of samples. Caramalau et al. [53] proposed an AL framework constructed on a sequential GCN, which can learn the similarity representation of strongly correlated samples, combined it with the uncertainty-based theory and CoreSet (i.e., UncertainGCN and CoreGCN) to query the unlabeled samples.

Although methods based on AL theory have a wide range of options for traditional images, AL methods applied to HSIC tasks still only use uncertainty-based theory mostly. For example, Li [54] proposed a method that measures the uncertainty of a given sample by the difference between two maximum predicted results of the model. Sun et al. [55] proposed a computational efficiency AL (CEAL) algorithm, which utilizes a clustering algorithm to preassign group labels for each unlabeled sample and query the samples with the highest uncertainty from each cluster. However, HSIs are prone to the phenomenon of foreign matters with the same spectrum and the same substance having different spectrums, resulting in dense spatial distribution between different categories. In addition, the category center is difficult to determine with a close spatial distribution. However, the number of unlabeled samples that obtained different predicted results by the same classification model trained with limited labeled samples is incalculable. Inaccurate category center selection also will cause inaccurate category measuring. Various reasons influenced the sampling effect of AL based-methods for HSI and affect the accuracy of HSI classification.

In this article, a new AL-based method for HSI classification based on prototype learning (PL) and GCN (called PLGCN) is proposed to efficiently select the highly valuable and informative samples from unlabeled samples in the candidate set for manual annotating and expanding the limited labeled sample set, thereby improving the accuracy of HSI classification. First, in the process of iteratively selecting the samples, updating the prototypes (i.e., the mean distance center of each class of samples) of each class in the limited labeled sample set. It will alleviate the problem that the class center is difficult to distinguish. Second, according to the characteristic of GCN that can transfer information between sample nodes and learn the similarity between strongly related samples, GCN layers are utilized for optimizing the spatial distribution among unlabeled samples. Finally, combined unlabeled features after optimizing by GCN layers with PL theory for adaptive excavating core samples to manual annotation. The PLGCN can adapt to different HSIC models, the effectiveness of which is verified in three benchmark datasets and exhibited the effect with different classical HSIC models.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:16:59 UTC from IEEE Xplore. Restrictions apply.

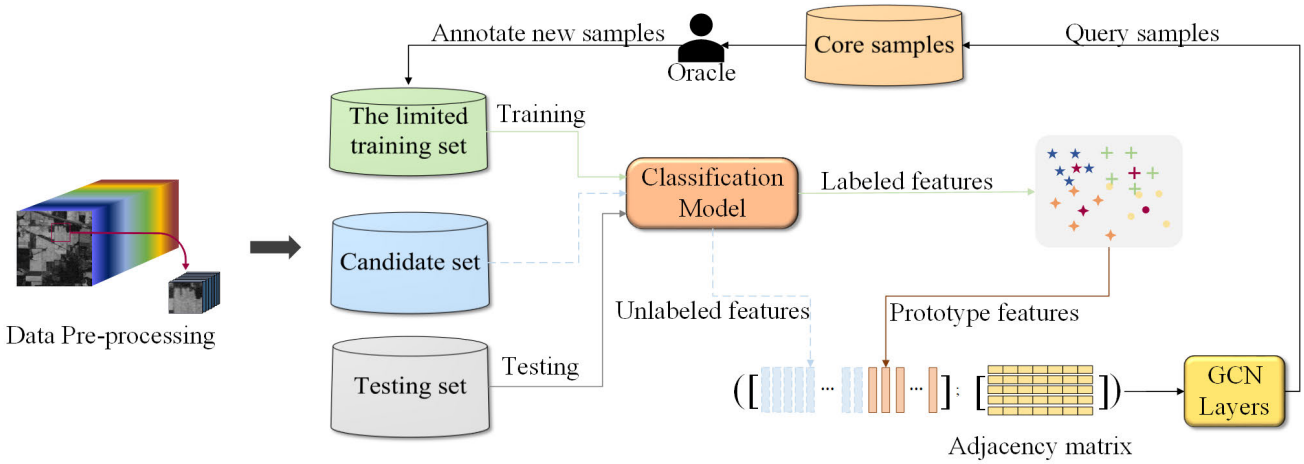



Fig. 1. Diagram outlines the proposed method.

The main contributions of this article can be summarized as follows.

1) To solve the problem that it is difficult to determine the class center between samples with tight spatial distribution, updating the prototype of each category in the expanding labeled sample set and exploring the similarities and differences between unlabeled samples and labeled samples for heightening the effect of evaluating unlabeled samples.
2) Based on the characteristic that GCN can transfer information between nodes, the spatial distribution of unlabeled samples is optimized by combining with prototypes of each category, thus enhancing the sampling effect.
3) Compared with the existing classical AL methods, the experimental results obtained on three benchmark HSI datasets by adapting to the different classical HSIC models state the proposed method provides competitive performance advantages.

## II. Proposed Method

The overall process of applying PLGCN is described in Fig. 1. In the first step, the HSI data need to be preprocessed. We extract the neighborhood data cubes centered on each pixel in HSI and process the data with different disposal according to whether the HSIC method needs dimension reduction processing. And we divide all the extracted data cubes into three sets with different functions, including the limited training set, candidate set, and testing set. In addition, the classification model is trained by limited labeled samples from the training set, we utilize the trained model to obtain embedding features of samples in the training set and candidate set. And we obtain the prototype of each category of embedded features in the training set, especially, the prototypes will be updated after each round of sampling operation. In the second step, the prototypes from the training set and unlabeled embedding features from the candidate set will be input into GCN for optimizing the spatial distribution of unlabeled embedding features. And we adaptively select core samples according to the prototype features and the optimized features. Finally, the training set is augmented by the selected core samples and utilized for continuing training of the model. Simultaneously, the testing set is utilized for assessing the performance of the method at different sampling periods. The process in detail of the sampling strategy indicates in Sections II-B–II-D.

### A. Data Preprocessing

Commonly, the HSI is denoted as $R^{H\times W\times B}$, the height and width of HSI are indicated by $H$ and $W$, respectively, and $B$ represents the spectral dimension of HSI. To extract the neighborhood data cube of each pixel in HSI, and ensure that each independent extracted data cube owns the same spatial dimension $P \times P$. For this reason, we foremost need to expand the dimension along the edge of HSI in different directions. The size of expanded rows or columns is related to the spatial size of extracted data cubes, which can be expressed as $(P-1)/2$. All the extracted data cubes compose the original dataset $\{U_i\}_{i=1}^{N}$, and $N$ is indicated that the number of all the pixels in HSI. According to whether the HSIC framework requires dimension reduction operation for adapting the design of different methods, it can be divided into the following two preprocessing patterns.

1) Extracting $P \times P$-neighborhood data cubes centered on each pixel from $R^{H\times W\times B}$ with the full spectral dimension $B$.
2) Extracting data cubes with each pixel as the center from $R^{H\times W\times B}$ after dimension reduction processing. In order to improve the calculation efficiency of the model, PCA is utilized for reducing the data redundancy of HSI before data cube extraction. After dimension reduction, the height and width of HSI will remain constant, and the spectral dimension will be changed to $D$. The spectral dimension of each extracted data cube will be consistent with HSI after dimension reduction.

When the preprocessing operation has been finished, we divide the original dataset $\{U_i\}_{i=1}^{N}$ into the training set $\{P_j\}_{j=1}^{M}$, candidate set $\{C_u\}_{u=1}^{Q}$, and testing set $\{T_t\}_{t=1}^{S}$, where, $M$ denotes the quantity of the training set, $Q$ denotes the quantity of the candidate set and $S$ denotes the quantity of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:16:59 UTC from IEEE Xplore. Restrictions apply.

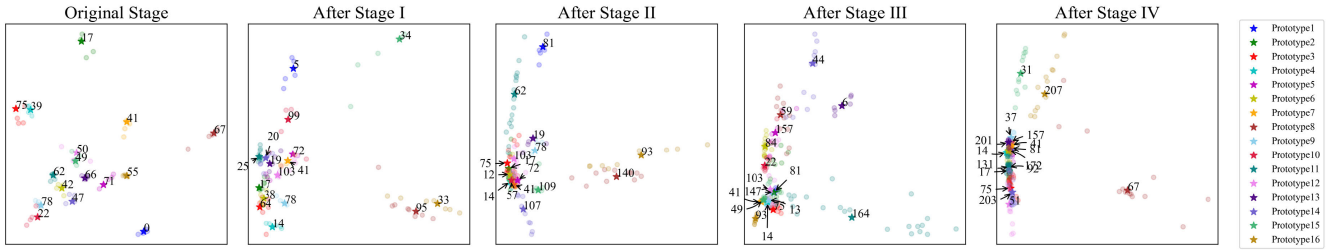



Fig. 2. Embedding feature distribution of training set in different sampling stages by the sample of Indian Pines dataset.

the testing set. The training set includes a modest amount of the labeled samples for training the HSIC model initially. The candidate set contains multitudinous unlabeled samples for extracting core samples by AL method, and the testing set has a similar quantity to the candidate set for testing. Samples in the training set, candidate set, and testing set are nonoverlapping.

### B. Prototype Iteratively Updating

PL aims to seek a prototype set to reflect the target set so that the training sample space can be reduced and the availability of data can be improved. Currently, there are two types of prototype acquisition: one is to directly select representative samples from the source set as prototypes, and the other is to generate prototypes in the source set through data fusion. In this article, consideration based on enhancing the effect of sample selection, we obtain the prototypes of each category of samples in the training set through the first acquisition way described above.

Before the prototypes obtaining, the HSIC model is trained by samples $\{P_j\}_{j=1}^{M}$ of training set. Then, the trained model $F_\theta$ maps samples $\{P_j\}_{j=1}^{M}$ to an embedding space of outputs $y \in Y$ ($Y$ represents the target space). Therefore, we obtain the embedding features $\{E_j\}_{j=1}^{M}$ of samples $\{P_j\}_{j=1}^{M}$, and continue prototype acquisition. In [56], points in embedding space gather around the prototype of each class $k$. Here, the prototype of each class is defined as the distance mean center sample of each class in the embedding space. The specific steps of prototype acquisition are as follows.

*Step 1:* Calculating the Euclidean distance between the discriminate vector of each sample and the other samples in class $k$. The Euclidean distance $\text{dist}_{(\alpha,\beta)}$ between the $\alpha$th discriminate vector and the $\beta$th one can be formulated as

$$\text{dist}_{(\alpha,\beta)} = \left\| E_\alpha - E_\beta \right\|_2. \tag{1}$$

Based on this, the distance set $\{D_\mu\}_{\mu=1}^{L_k}$ from each embedding vector to the other in $k$-class can be obtained. $L_k$ denotes as the quantity of such distance vector in $k$-class.

*Step 2:* For obtaining the prototype of each class $p_k$, the objective is to minimize all the average of the distance vectors $\{D_\mu\}_{\mu=1}^{L_k}$ in that class. The objective defined as

$$\text{aver}_\mu = \frac{1}{L_k} \sum_{\mu=1}^{L_k} \text{dist}_\mu \tag{2}$$

where, $\text{aver}_\mu$ represents the average of all the distance elements $\text{dist}_\mu$ in the $\mu$th distance vector of distance set $\{D_\mu\}_{\mu=1}^{L_k}$. These obtained average elements will form a set $\{\text{aver}_\mu\}_{\mu=1}^{L_k}$

$$p_k = \arg\min\left\{\text{aver}_\mu\right\}_{\mu=1}^{L_k}. \tag{3}$$

So far, we obtained the prototype set $\{D_k\}_{k=1}^{K}$, $K$ represents the number of classes in HSI.

In particular, as the training set is expanded, the distribution of the training set will be changed, so the prototypes will be updated after each stage of the sampling operation. The change in embedding feature distribution of the training set in different sampling stages by the sample of Indian Pines dataset is shown in Fig. 2.

### C. Spatial Distribution of Features Optimizing via GCN

The information about each node in the graph and the relationship between the current node and its associated nodes can be learned through GCN, which can aggregate the characteristics of the current node and the associated nodes.

In the sampling strategy of PLGCN, GCN is sequentially trained by the features of labeled and unlabeled samples extracted from the trained model $F_\theta$. Our motivation in utilizing GCN layers is primarily in optimizing the spatial distribution of unlabeled samples through learning the differences in the representations between prototypes and unlabeled samples in embedding space. And we utilize GCN as a binary classifier to distinguish which unlabeled samples are annotated.

A graph $\mathcal{G}$ is the input of GCN, which consists of vertices $\mathcal{V}$ (i.e., entered embedding features) and edges $\mathcal{E}$ (i.e., the relationship between the vertices). Vertices $\mathcal{V}$ are composed of prototypes and unlabeled features, where edges information is encoded in an adjacency matrix $A$. The initial elements of an adjacency matrix $A$ involve the vector product between each embedding feature of $\mathcal{V}$ (i.e., $S_{\delta\xi} = \mathcal{V}_\delta^\top \mathcal{V}_\xi$, $\{\delta, \xi\} \in L_\mathcal{V}$, $L_\mathcal{V}$ represents the addition result of the number of unlabeled samples and prototypes) after applying the $l_2$ normalization. And then the whole encoding process can be simply expressed as

$$A = D^{-1}(S - 1) + I \tag{4}$$

where, $D$ signifies the degree of $S$ and $I$ is the identify matrix.

For fear of features obtained oversmoothing via GCN [57], we design a two-layer architecture. The operation course of the first GCN layer can be expressed as

$$X^1 = F_g^1(A, \mathcal{V}; \Theta_1) \tag{5}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:16:59 UTC from IEEE Xplore. Restrictions apply.

where, $X^1 \in \mathbb{R}^{h \times L_\mathcal{V}}$ represents the output feature of the first GCN layer, $h$ represents the number of hidden layer units, and $\Theta_1$ are the parameters of the first GCN layer.

We deploy a rectified linear unit activation (ReLU) [58] after the first GCN layer, which can promote gradient descent more effectively. The objective of utilizing GCN layers is to judge whether the sample is labeled, thus, we assign pseudolabel 1 to the labeled embedding features and pseudolabel 0 to the unlabeled features. And the output of the second GCN layer is activated via a sigmoid function. The whole network function can be further defined as

$$F_g = \sigma(\Theta_2(\mathrm{ReLU}(\Theta_1 A))A) \tag{6}$$

where, $\Theta_2$ are the parameters of the second GCN layer.

To achieve this object, we deploy a cross-entropy function as our loss function. It is defined as

$$L_g = -\frac{1}{K}\sum_{c=1}^{K}\log(F_g(\mathcal{V}, A)_c) - \frac{1}{L_v - L_{vl}}\sum_{c=L_{vl}+1}^{L_v}(1 - \log(F_g(\mathcal{V}, A)_c)). \tag{7}$$

Through the above calculation operation, we obtain the optimized unlabeled features $\{G_u\}_{u=1}^{Q}$ and the optimized prototypes $\{G_k\}_{k=1}^{K}$ in another embedding space.

### D. Selecting Core Samples

After we obtain the optimized features via GCN, core samples are selected by the optimized prototypes and unlabeled embedding features. This process is shown in Fig. 3.

The distance between each optimized unlabeled feature and each optimized prototype feature is expressed as

$$\mathrm{dist}_{(G_u, G_k)} = \left\| G_u - G_k \right\|_2 \tag{8}$$

where, $G_u$ represents the current optimized unlabeled feature and $G_k$ represents the prototype of the $k$-class. And we calculate the distance between each unlabeled sample feature and each class of prototype. For each unlabeled feature, we obtain a set of distances $\{\mathrm{dist_{up}}\}_{\mathrm{up}=1}^{K}$ from each prototype.

For each distance set $\{\mathrm{dist_{up}}\}_{\mathrm{up}=1}^{K}$, we divide the unlabeled feature $G_u$ nearest to the prototype $G_k$ into $k$-category. This operation is expressed as

$$k = \arg\min\{\mathrm{dist_{up}}\}_{\mathrm{up}=1}^{K}. \tag{9}$$

For each category, we obtain a set of distances $\{\mathrm{dist_{pu}}\}_{\mathrm{pu}=1}^{Q_k}$ between the current prototype and each unlabeled feature in the current class. $Q_k$ represents the number of unlabeled features divided into $k$-class.

And then we select the nearest and farthest unlabeled samples from the prototype in $k$-category so that we can ensure the selection performance of edge samples and center samples. These selected samples are namely core samples. This process can be shown in

$$P = P \bigcup \arg\min\{\mathrm{dist_{pu}}\}_{\mathrm{pu}=1}^{Q_k} \tag{10}$$

$$P = P \bigcup \arg\max\{\mathrm{dist_{pu}}\}_{\mathrm{pu}=1}^{Q_k}. \tag{11}$$

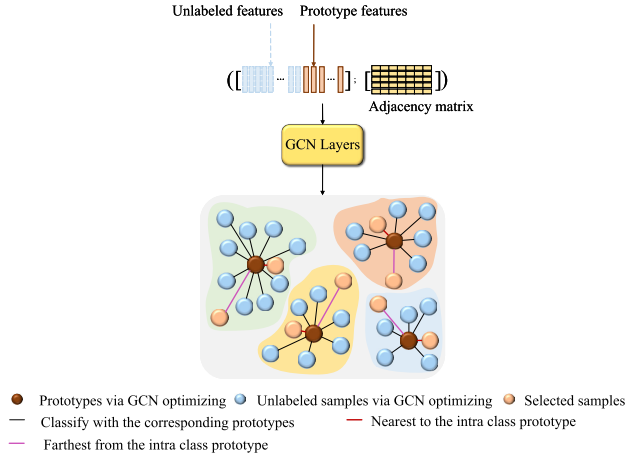



Fig. 3. Schematic of core sample selection after GCN optimizing.

**Algorithm 1** PLGCN

**Input:** Initial labeled dataset $D_P$, candidate dataset $D_C$ and the sampling period $b$.

Initialize $(\boldsymbol{x_P}, \boldsymbol{y_P})$ and $(\boldsymbol{x_C})$ - data and labels for labeled samples, data for candidate samples.
**for** $\boldsymbol{i = 1 \rightarrow b}$ **do**
  $\boldsymbol{\theta \leftarrow f(x_P, y_P)}$ *Training classification model with labeled data*
  Obtaining prototype set $\boldsymbol{D_k}$ via (1)-(3)
  $\boldsymbol{\mathcal{V} = [v_{D_k}, v_C] \leftarrow f(x_{D_k}, x_C; \theta)}$ *Extracting features for prototypes and candidate data*
  Calculating adjacency matrix $\boldsymbol{A}$ via (4)
  $\boldsymbol{\Theta \leftarrow f_g(\mathcal{V}, A)}$ *Training the GCN*
  $\boldsymbol{G_k, G_C \leftarrow f_g(\mathcal{V}, A; \Theta)}$ *Extracting optimized features for prototypes and candidate data*
  Obtaining core samples set $\boldsymbol{D_{core}}$ via (8)–(11)
  $\boldsymbol{D_P = D_P \bigcup D_{core}}$ *Adding core sample set to labeled sample set*
**end for**

After the sampling work is completed, we will iterate this process from Sections II-B–II-D until the sampling period $b$ we set is satisfied. And we summarize the process of the PLGCN as Algorithm 1.

## III. Experiments and Analysis

In this chapter, we introduce the experimental HSI datasets and the experimental configuration in detail. At the same time, we reveal the analysis of the comparative experimental results in this section.

### A. Data Description

In this article, comparative experiments are settled in three common his datasets. The experimenthisHSI datasets in detail as follows.

*1) Indian Pines Dataset:* The Indian Pines dataset (abbreviated as IP) contains a HSI, collected from the scene over the Indian Pines test collected from the scene over the Indian Pines test site in North-western Indiana through the AVIRIS sensor.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:16:59 UTC from IEEE Xplore. Restrictions apply.

TABLE I
LAND COVER CLASSES, NUMBER OF TRAINING SAMPLES, NUMBER OF SAMPLES IN CANDIDATE SET, AND NUMBER OF TESTING SAMPLES FOR THE IP DATASET

| Class No. | Class Name | Total | Training Samples | Samples in candidate set | Testing samples |
|---|---|---|---|---|---|
| 1 | Alfalfa | 46 | 5 | 19 | 22 |
| 2 | Corn-notill | 1428 | 5 | 711 | 712 |
| 3 | Corn-mintill | 830 | 5 | 398 | 427 |
| 4 | Corn | 237 | 5 | 113 | 119 |
| 5 | Grass-pasture | 483 | 5 | 238 | 240 |
| 6 | Grasstrees | 730 | 5 | 348 | 377 |
| 7 | Grass-pasture-mowed | 28 | 5 | 17 | 6 |
| 8 | Background | 478 | 5 | 253 | 220 |
| 9 | Oats | 20 | 5 | 8 | 7 |
| 10 | Soybean-notill | 972 | 5 | 491 | 476 |
| 11 | Soybean-mintill | 2455 | 5 | 1421 | 1209 |
| 12 | Soybean-clean | 593 | 5 | 297 | 291 |
| 13 | Wheat | 205 | 5 | 106 | 94 |
| 14 | Woods | 1265 | 5 | 623 | 637 |
| 15 | Buildings-grass-trees-drives | 386 | 5 | 184 | 197 |
| 16 | Stone-steel-towers | 93 | 5 | 37 | 51 |
| Total | | 10249 | 80 | 5084 | 5085 |

The spatial size of this HSI is 145 × 145 and the amount of spectral reflectance bands is 224. The wavelength range of all spectral reflectance bands is between $0.4-2.5 \times 10^{-6}$ meters. The creators of the dataset also further optimized the dataset, and they removed the bands covering the region of water absorption and the dimension of the final data is 200. In addition to the background, the available pixels can be divided into 16 categories and the total number of available pixels is 10 249. Among them, the number of samples for different purposes is shown in Table I.

*2) Salinas Dataset:* The Salinas dataset (abbreviated as SA) contains a HSI, which gathered the scene over Salinas Valley in California through the AVIRIS sensor. It has a high spatial resolution that 3.7-m pixels. The spatial size of this HSI is 512 × 217 and the amount of spectral reflectance bands is 224. 20 water absorption bands are divested so that the final dimension of this dataset is 204. In addition to the background, the available pixels can be divided into 16 categories and the total number of available pixels is 54 129. Among them, the number of samples for different purposes is shown in Table II.

*3) University of Pavia Dataset:* The University of Pavia Dataset (abbreviated as PU) contains a HSI, which took the scene over Pavia University in northern Italy through the ROSIS sensor. It covered the area with the spatial size 610 × 340 and the spectral dimension 103. In addition to the background, the available pixels can be divided into nine categories and the total number of available pixels is 42 776. Among them, the number of samples for different purposes is shown in Table III.

### B. Experimental Configuration

In this article, all the comparative experiments are handled on a desktop computer with 12th Gen Intel[1] Core[2] i7-12700F 2.10 GHz CPU and the Nvidia RTX 3060 TI GPU. And the codes were debugged in Windows 11 environment. Pytorch 1.12.0 and Python 3.8.13 are utilized for the implementation of deep learning frameworks.

[1]Registered trademark.
[2]Trademarked.

In our experiments, we designed a two-layer GCN and employed Adam as an optimizer with a learning rate settled at 0.001. We trained the GCN with different training epochs for each HSI dataset and the number of hidden units is the same as the weight of features extracted from the classification model. The training epoch number of GCN for each HSI dataset is shown in Table IV. The number of hidden units of GCN from different classification model is shown in Table V. And the number of sampling cycles is set to 5 (the cycle of the compared methods is the same).

The applied HSI classification models are 3-D-CNN [28], HybridSN [32], SSRN [37], attention-based adaptive spectral–spatial kernel ResNet ($A^2S^2K$-ResNet) [59], and Integrates hybrid pyramid feature fusion and coordinate attention network (HPCAN) [40]. The frameworks of these models in detail are shown as follows.

*1) 3-D-CNN:* The framework of 3-D-CNN contains two 3-D-convolution layers with ReLU as the activation. The convolution kernel size of 3-D-convolution layers is 3 × 3 × 3. And it has two pooling layers and the whole structure of 3-D-CNN by the example of IP dataset is shown in Fig. 4.

*2) HybridSN:* The framework of HybridSN contains three 3-D-convolution layers with different size of convolution kernels. The sizes of different convolution kernels include 3 × 3 × 7, 3 × 3 × 5, 3 × 3 × 3. A 2-D-convolution layer is connected behind these above 2-D-convolution and 3-D-convolution layers. The size of layer is 3 × 3. And it has three fully connected layers (i.e., linear layer) and the whole structure of HybridSN by the example of IP dataset is shown in Fig. 5.

*3) SSRN:* The framework of SSRN contains two spectral residual modules with skip connection and two spatial residual modules with skip connection. A spectral residual module contains two 3-D-convolution layers with kernel size of 1 × 1 × 11. A spatial residual module contains two 3-D-convolution layers with kernel size of 3 × 3. And behind these layers, it has a pooling layer and a fully connected layer. The whole structure of SSRN by the example of the IP dataset is shown in Fig. 6.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:16:59 UTC from IEEE Xplore. Restrictions apply.

TABLE II
LAND COVER CLASSES, NUMBER OF TRAINING SAMPLES, NUMBER OF SAMPLES IN CANDIDATE SET, AND NUMBER OF TESTING SAMPLES FOR THE SA DATASET

| Class No. | Class Name | Total | Training Samples | Samples in candidate set | Testing samples |
|---|---|---|---|---|---|
| 1 | Brocoli_green_weeds_1 | 2009 | 5 | 998 | 1006 |
| 2 | Brocoli_green_weeds_2 | 3726 | 5 | 1853 | 1868 |
| 3 | Fallow | 1976 | 5 | 960 | 1011 |
| 4 | Fallow_rough_plow | 1394 | 5 | 666 | 723 |
| 5 | Fallow_smooth | 2678 | 5 | 1338 | 1335 |
| 6 | Stubble | 3959 | 5 | 1954 | 2000 |
| 7 | Celery | 3579 | 5 | 1760 | 1814 |
| 8 | Grapes_untrained | 11271 | 5 | 5618 | 5648 |
| 9 | Soil_vineyard_develop | 6203 | 5 | 3152 | 3046 |
| 10 | Corn_senesced_green weeds | 3278 | 5 | 1635 | 1638 |
| 11 | Lettuce_romaine_4wkl | 1068 | 5 | 563 | 500 |
| 12 | Lettuce_romaine_5wkl | 1927 | 5 | 920 | 1002 |
| 13 | Lettuce_romaine_6wkl | 916 | 5 | 434 | 477 |
| 14 | Lettuce_romaine_7wkl | 1070 | 5 | 549 | 516 |
| 15 | Vineyard_untrained | 7268 | 5 | 3719 | 3544 |
| 16 | Vineyard_vertical_trellis | 1807 | 5 | 905 | 897 |
| Total | | 54129 | 80 | 27024 | 27025 |

TABLE III
LAND COVER CLASSES, NUMBER OF TRAINING SAMPLES, NUMBER OF SAMPLES IN CANDIDATE SET, AND NUMBER OF TESTING SAMPLES FOR THE PU DATASET

| Class No. | Class Name | Total | Training Samples | Samples in candidate set | Testing samples |
|---|---|---|---|---|---|
| 1 | Asphalt | 6631 | 5 | 3293 | 3333 |
| 2 | Meadows | 18649 | 5 | 9306 | 9338 |
| 3 | Gravel | 2099 | 5 | 1041 | 1053 |
| 4 | Trees | 3064 | 5 | 1541 | 1518 |
| 5 | Painted metal sheets | 1345 | 5 | 670 | 670 |
| 6 | Bare soil | 5029 | 5 | 2532 | 2492 |
| 7 | Bitumen | 1330 | 5 | 679 | 646 |
| 8 | Self-blocking bricks | 3682 | 5 | 1830 | 1847 |
| 9 | Shadows | 947 | 5 | 473 | 469 |
| Total | | 42776 | 80 | 21365 | 21366 |

TABLE IV
TRAINING EPOCHS OF GCN FOR EACH HSI DATASET

| Datasets | Training Epochs of GCN |
|---|---|
| IP | 50 |
| SA | 70 |
| PU | 50 |

TABLE V
NUMBER OF HIDDEN UNITS OF GCN FROM DIFFERENT CLASSIFICATION MODEL

| Model | The Number of Hidden Units of GCN |
|---|---|
| 3D-CNN | 384 |
| HybridSN | 128 |
| SSRN | 28 |
| $A^2S^2K$-ResNet | 480 |
| HPCAN | 192 |

*4) $A^2S^2K$-ResNet:* Different from the above three models, the spectral–spatial channel attention is introduced for enhancing the differentiate ability of model to identify key factors. Besides, $A^2S^2K$ layer is utilized for learning the selection of spectral–spatial kernel. The framework of $A^2S^2K$-ResNet contains four Res Blocks and pooling layer. The whole structure of $A^2S^2K$-ResNet by the example of the IP dataset is shown in Fig. 7.

*5) HPCAN:* Coordinate attention is introduced for enhancing perception of pixel position and direction of model. And hybrid pyramid feature fusion is utilized through 3-D-convolution layers with size of $3 \times 3 \times 3$, 2-D-convolution layers with size of $1 \times 1$ and $3 \times 3$, Res Blocks. The whole structure of HPCAN by the example of the IP dataset is shown in Fig. 8.

Table VI shows the settings of training epochs and the learning rate of optimizers for applied HSIC models in this article, respectively. The above settings are based on the code settings in relevant papers. Table VII shows the spatial size of the input data cubes for different models on different datasets, and corresponding preprocessing strategies are set for dimensionality reduction operations in the original design for different models. In the network architecture of 3-D-CNN, HybridSN, and HPCAN, it is necessary to reduce the dimensionality of input data cubes to ensure the computational efficiency of the models. Therefore, the same settings are maintained in this article. Overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa) are used to estimate the expression of different methods. OA is an indicator to evaluate the proportion of the amount of correctly classified pixels in the total amount of pixels. AA is an indicator to evaluate the AA of all categories. Kappa is an indicator to measure the effect of model classification.

### C. Ablation Study

As described in Section II, the proposed PLGCN involves two critical components to ensure the completeness of sample

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:16:59 UTC from IEEE Xplore. Restrictions apply.

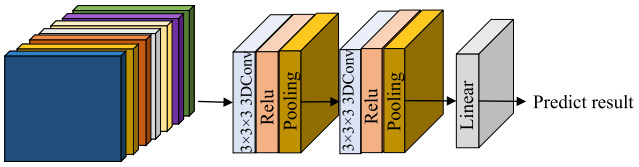



Fig. 4. Structure diagram of 3-D-CNN by the example of IP dataset.

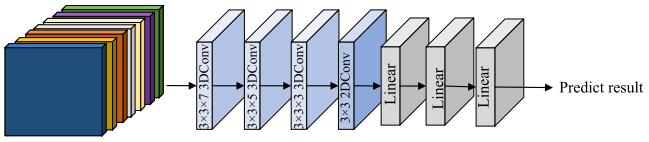



Fig. 5. Structure diagram of HybridSN by the example of IP dataset.

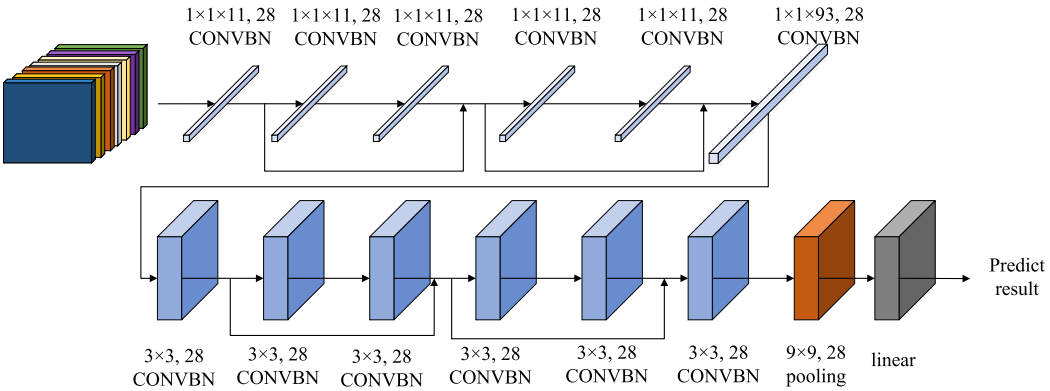



Fig. 6. Structure diagram of SSRN by the example of IP dataset.

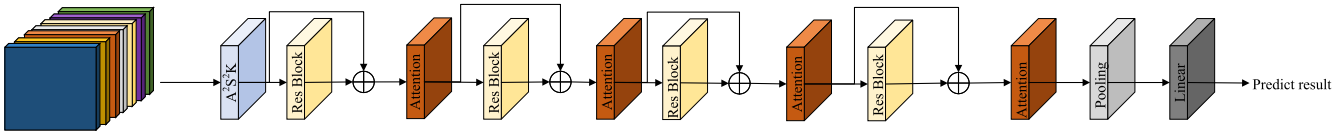



Fig. 7. Structure diagram of $A^2S^2K$-ResNet by the example of IP dataset.

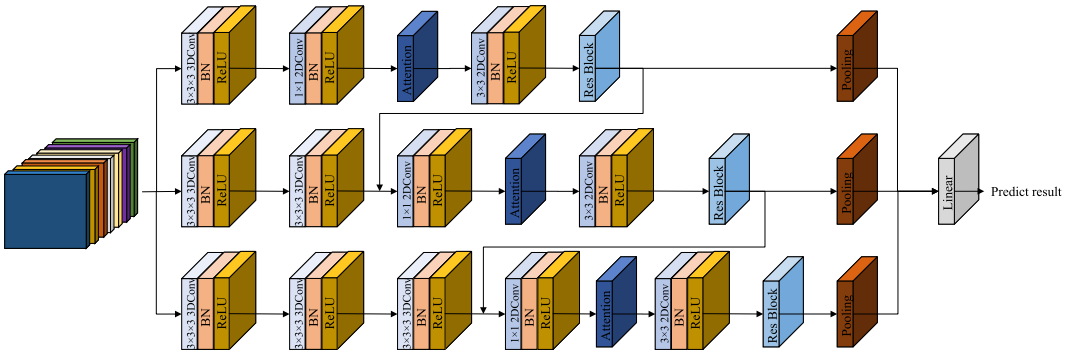



Fig. 8. Structure diagram of HPCAN by the example of IP dataset.

sampling and enhance the classification effect, i.e., iterative updating of prototypes for each class of training samples at each sampling stage and spatial distribution of unlabeled sample features optimizing via GCN. An ablation study was designed on three datasets to reveal the impact of these two parts. Specifically, we evaluate the original classification performance of five baselines (include 3-D-CNN, HybridSN, SSRN, $A^2S^2K$-ResNet, and HPCAN) and the classification performance after gradually implementing the PL and GCN. PL means iteratively updating the prototypes at each AL operation stage, and selecting unlabeled samples to expand the training set based on the determined prototype. It is notable that the collaborative effect of prototypes is indispensable in the process of optimizing unlabeled sample features through GCN. Therefore, no separate experiments with GCN-only additions were conducted.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:16:59 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
TRAINING PARAMETERS OF THE APPLIED HSI CLASSIFICATION MODELS

| Model | Epochs | Learning Rate |
|---|---|---|
| 3D-CNN | 200 | 0.001 |
| HybridSN | 150 | 0.001 |
| SSRN | 150 | 0.001 |
| $A^2S^2K$-ResNet | 150 | 0.001 |
| HPCAN | 150 | 0.002 |

TABLE VII
DETAIL OF THE INPUT FOR THE APPLIED MODELS ON DIFFERENT DATASETS

| Model | Input Size | | | Dimension |
|---|---|---|---|---|
| | IP | SA | PU | Reduction |
| 3D-CNN | $15 \times 15$ | $19 \times 19$ | $19 \times 19$ | Yes |
| HybridSN | $15 \times 15$ | $19 \times 19$ | $19 \times 19$ | Yes |
| SSRN | $11 \times 11$ | $11 \times 11$ | $11 \times 11$ | No |
| $A^2S^2K$-ResNet | $11 \times 11$ | $11 \times 11$ | $11 \times 11$ | No |
| HPCAN | $15 \times 15$ | $19 \times 19$ | $19 \times 19$ | Yes |

TABLE VIII
ABLATION EXPERIMENT RESULTS ON IP DATASET

| Method | OA | AA | Kappa |
|---|---|---|---|
| 3D-CNN | 57.66 | 73.92 | 53.87 |
| 3D-CNN + Prototype learning | 69.03 | 79.42 | 64.77 |
| 3D-CNN + Prototype learning + GCN | **78.25** | **86.09** | **75.29** |
| HybridSN | 56.20 | 68.70 | 51.48 |
| HybridSN + Prototype learning | 70.54 | 80.27 | 66.57 |
| HybridSN + Prototype learning + GCN | **80.10** | **84.50** | **79.57** |
| SSRN | 70.72 | 82.98 | 67.20 |
| SSRN + Prototype learning | 72.30 | 80.95 | 67.29 |
| SSRN + Prototype learning + GCN | **80.22** | **87.81** | **78.07** |
| $A^2S^2K$-ResNet | 46.00 | 60.00 | 39.34 |
| $A^2S^2K$-ResNet + Prototype learning | 55.24 | 69.11 | 49.40 |
| $A^2S^2K$-ResNet + Prototype learning + GCN | **63.70** | **69.94** | **58.99** |
| HPCAN | 75.54 | 86.15 | 72.63 |
| HPCAN + Prototype learning | 80.51 | 85.52 | 77.84 |
| HPCAN + Prototype learning + GCN | **91.05** | **94.26** | **89.77** |

TABLE IX
ABLATION EXPERIMENT RESULTS ON SA DATASET

| Method | OA | AA | Kappa |
|---|---|---|---|
| 3D-CNN | 90.29 | 89.27 | 89.25 |
| 3D-CNN + Prototype learning | 92.86 | 94.59 | 92.06 |
| 3D-CNN + Prototype learning + GCN | **97.91** | **98.23** | **97.68** |
| HybridSN | 92.84 | 95.99 | 92.05 |
| HybridSN + Prototype learning | 94.94 | 96.90 | 94.37 |
| HybridSN + Prototype learning + GCN | **97.96** | **98.18** | **97.73** |
| SSRN | 93.15 | 96.23 | 92.74 |
| SSRN + Prototype learning | 95.47 | 97.02 | 94.86 |
| SSRN + Prototype learning + GCN | **97.67** | **97.81** | **96.95** |
| $A^2S^2K$-ResNet | 86.18 | 92.33 | 85.92 |
| $A^2S^2K$-ResNet + Prototype learning | 89.58 | 93.25 | 88.25 |
| $A^2S^2K$-ResNet + Prototype learning + GCN | **94.81** | **96.82** | **93.96** |
| HPCAN | 94.20 | 96.63 | 93.57 |
| HPCAN + Prototype learning | 95.70 | 97.68 | 95.22 |
| HPCAN + Prototype learning + GCN | **98.05** | **98.26** | **97.83** |

TABLE X
ABLATION EXPERIMENT RESULTS ON PU DATASET

| Method | OA | AA | Kappa |
|---|---|---|---|
| 3D-CNN | 67.74 | 72.51 | 59.81 |
| 3D-CNN + Prototype learning | 76.94 | 73.41 | 69.04 |
| 3D-CNN + Prototype learning + GCN | **87.19** | **78.69** | **82.62** |
| HybridSN | 66.46 | 60.19 | 57.83 |
| HybridSN + Prototype learning | 75.02 | 72.46 | 66.42 |
| HybridSN + Prototype learning + GCN | **83.30** | **71.02** | **77.59** |
| SSRN | 74.18 | 74.75 | 66.76 |
| SSRN + Prototype learning | 80.51 | 84.80 | 75.38 |
| SSRN + Prototype learning + GCN | **88.65** | **87.43** | **85.66** |
| $A^2S^2K$-ResNet | 59.27 | 59.53 | 48.38 |
| $A^2S^2K$-ResNet + Prototype learning | 66.16 | 65.54 | 57.32 |
| $A^2S^2K$-ResNet + Prototype learning + GCN | **79.02** | **73.94** | **71.65** |
| HPCAN | 65.45 | 74.67 | 58.30 |
| HPCAN + Prototype learning | 86.59 | 85.91 | 82.53 |
| HPCAN + Prototype learning + GCN | **94.38** | **90.14** | **92.51** |

Tables VIII–X exhibits ablation experiment results on the IP, SA, and PU datasets. First, by comparing the original classification performance of each baseline with the classification performance adopting the PL, it can be seen that the expansion of the finite training set through prototype iterative update gained results to some extent in improving the accuracy of HSI classification tasks. Especially for the evaluation and analysis of OA and KAPPA, the results of each baseline on different datasets have increased. However, the improvement in experiment indicators of minority models is relatively not obvious. After adopting the GCN to optimize the distribution of unlabeled embedding features in latent space, all indicators were significantly improved. Therefore, Learning the association between prototypes and unlabeled samples through GCN makes unlabeled sample features more discriminative after the spatial distribution is optimized, which is effective for sampling valuable samples to improve classification performance. PL and GCN work collaboratively to render satisfactory classification results.

### D. Results of Comparative Experiments

To validate the availability of the proposed method in HSIC tasks, we compare the proposed method with different types of AL methods which we mention below. The AL methods compared with the proposed method in this article include Random sampling, Entropy [42], CoreSet [44], BADGE [52], UncertainGCN [53], and CoreGCN [53]. Random sampling is the most elementary sampling technology. Entropy is an uncertainty-based method, which selects a batch of samples with the greatest uncertainty (i.e., maximum entropy). CoreSet is an AL method based on feature space coverage, which selects to add a group of unlabeled samples to minimize the maximum distance of other samples from the labeled sample set. BADGE integrates diversity and uncertainty, and the gradient normalization is used to indicate the degree of uncertainty. UncertainGCN and CoreGCN are GCN-based methods fused with the uncertainty-based theory and feature space coverage-based method, respectively. Our comparative

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:16:59 UTC from IEEE Xplore. Restrictions apply.

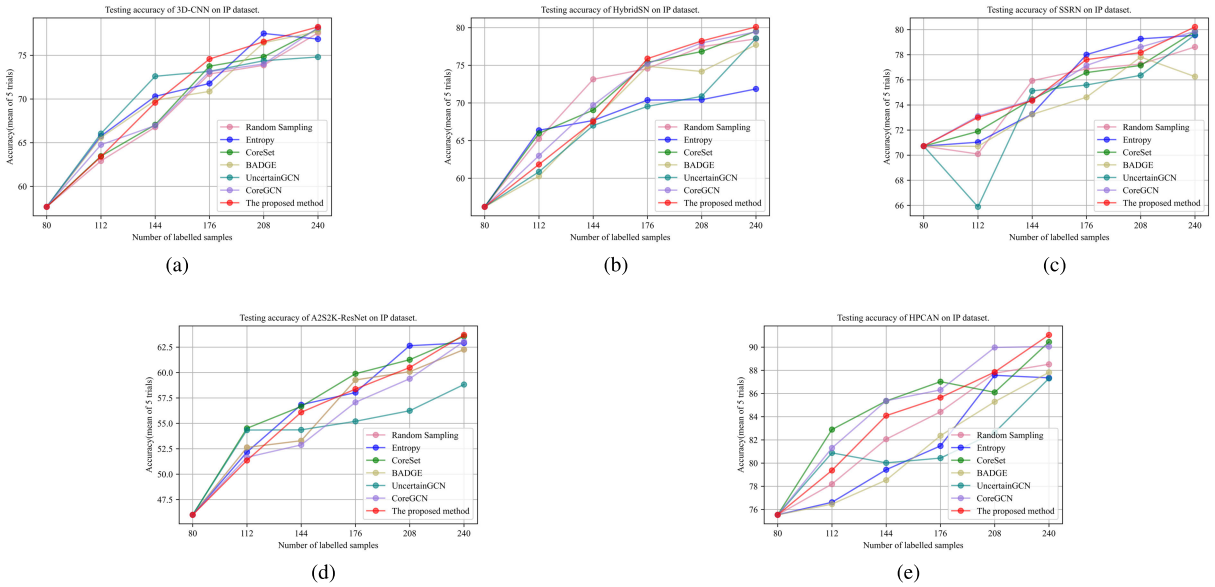



Fig. 9. Line chart of OA results by applying different classification models for different AL methods on IP dataset. (a) Applying 3-D-CNN as classification model, (b) applying HybridSN as classification model, (c) applying SSRN as classification model, (d) applying $A^2S^2K$-ResNet as classification model, and (e) applying HPCAN as classification model.

experiments are applied to different HSI classification models and carried out on three different datasets above introduced. In each sampling cycle, we sampled 32 unlabeled samples for each AL method on the IP dataset and SA dataset (the number of categories for both the IP dataset and SA dataset is 16) and 18 unlabeled samples on the PU dataset (the number of categories for PU dataset is 9).

Figs. 9–11 portray the changing trend of Classification OA by adapting different AL methods to different HSI classification models in different sampling stages on the IP dataset, SA dataset, and PU dataset. It can be seen from the line charts that the growth trend of applying PLGCN is stable compared with other methods. For example, the classification results of adapting BADGE method to the $A^2S^2K$-ResNet model show significant fluctuations in all three datasets. And applying Entropy and UncertainGCN perform erratically in the line charts of three datasets. In particular, the results of applying UncertainGCN in the IP dataset, Entropy in the SA dataset, and CoreSet in the PU dataset occurrence a downward trend in the first sampling stage. But the classification effects of adapting PLGCN to different HSIC models in three datasets show a steady upward trend. Although the PLGCN adapted to HSI classification models may not gain the best classification result in the first two stages, it gradually obtains the best result from the third stage and the results in the final stage are the best of all comparison methods. It shows that the sampling effect of applying PLGCN for training HSI classification models is stable and satisfactory.

Since the classification results of applying the HPCAN model are all the optimum in three datasets, we show the details of the results of different AL methods adapt to HPCAN below.

Tables XI–XIII describes classification results in detail after the final sampling stage by the example of adapting different AL methods to HPCAN on the IP dataset, SA dataset, and PU dataset. It can be seen from the tables that the OA, AA, and KAPPA of applying PLGCN express the best results in the IP datasets compared with other methods. The OA, AA, and KAPPA of using PLGCN are 0.61%, 0.59%, and 0.63% higher than the second-best results on the IP dataset, respectively. On the SA dataset and PU dataset, the OA and KAPPA of using PLGCN still perform optimally, which are 0.05% and 0.06% on the SA dataset, 1.66% and 2.23% on the PU dataset higher than the best results of other AL methods. Although the classification accuracy of each category of the proposed method is not always at the forefront, the overall effect is still outstanding and relatively high and close to the best result for these categories compared with other methods. It also can be seen that the sampling results of using PLGCN are more proportional and unlabeled samples in both classes with a large number of samples and classes with less quantity of samples all can be sampled.

Figs. 12–14 visualize the ground-truth maps of three HSI datasets, which are compared with the classification maps of the comparison experiments. Compared with the effect of using other AL methods to expand the limited training set, the results of using PLGCN exhibit fewer classification errors in most categories. Whether in categories with large or small sample sizes, the noise in the classification maps of PLGCN is relatively few, which reveals the superior sampling efforts of PLGCN for enhancing the classification accuracy of HSI classification models.

The PLGCN optimizes and adjusts the spatial distribution of HSI data through GCN, which alleviates the problem of unclear distribution among different types of samples of HSI data. Then, the samples are further adaptively selected with differences or close to the prototypes, which are updated iteratively to accurately select unlabeled samples that are more valuable for HSI classification model training. The comparative experiment proves the effectiveness of the proposed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:16:59 UTC from IEEE Xplore. Restrictions apply.

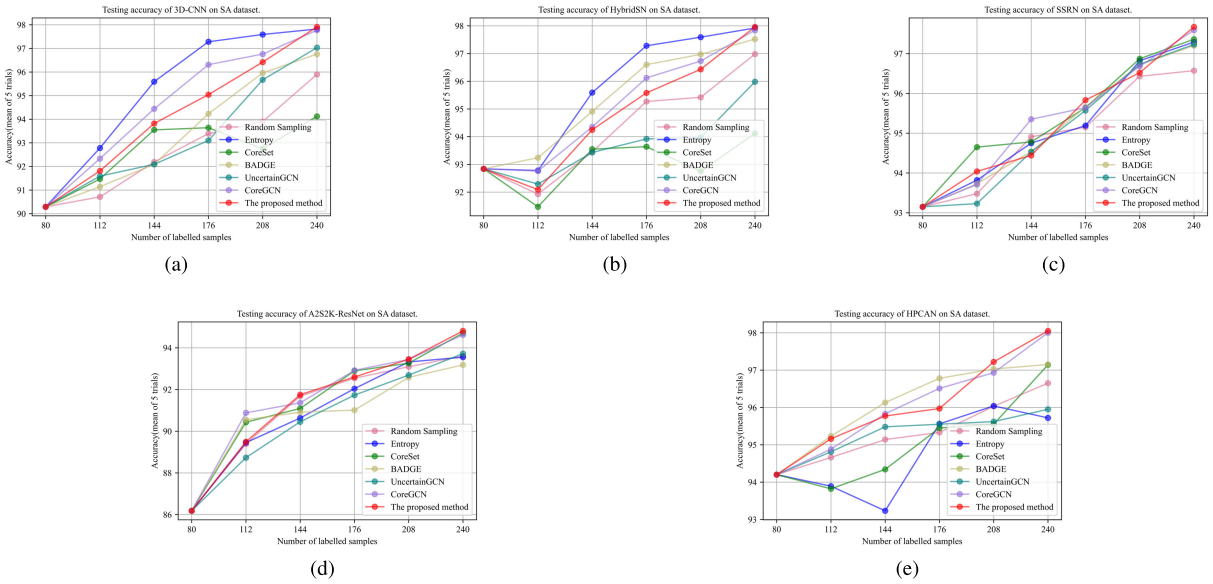



Fig. 10. Line chart of OA results by applying different classification models for different AL methods on SA dataset. (a) Applying 3-D-CNN as classification model, (b) applying HybridSN as classification model, (c) applying SSRN as classification model, (d) applying $A^2S^2K$-ResNet as classification model, and (e) applying HPCAN as classification model.

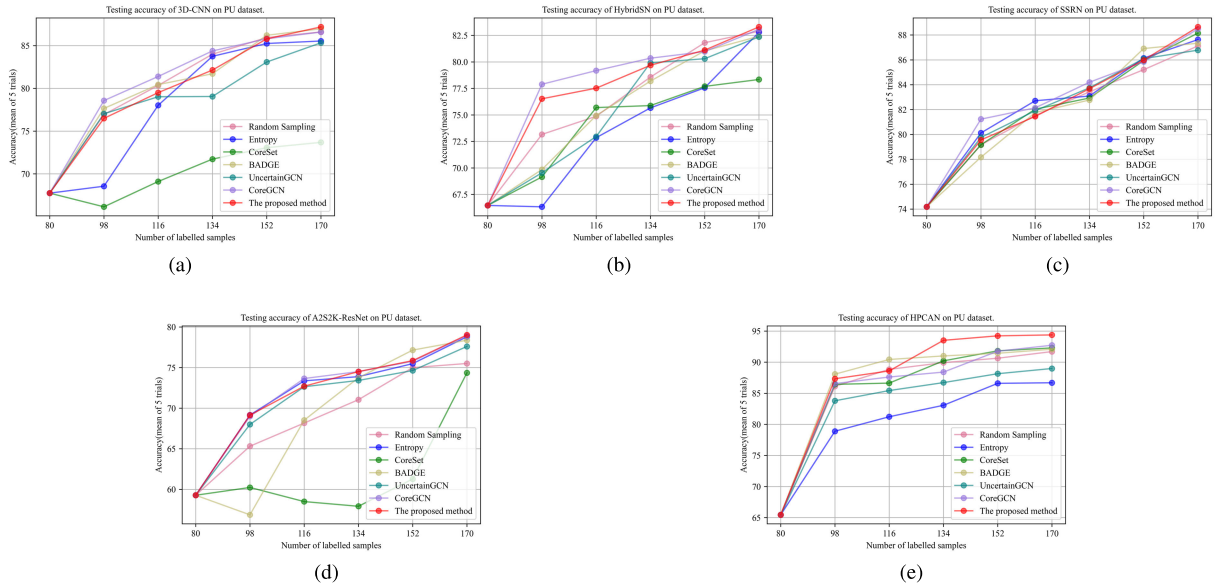



Fig. 11. Line chart of OA results by applying different classification models for different AL methods on PU dataset. (a) Applying 3-D-CNN as classification model, (b) applying HybridSN as classification model, (c) applying SSRN as classification model, (d) applying $A^2S^2K$-ResNet as classification model, and (e) applying HPCAN as classification model.

method for improving the classification accuracy of HSI classification models.

## IV. Discussion

### A. Influence of the Number of GCN Layers

In this article, we verified the influence of the different numbers of GCN layers on the sampling effect of PLGCN on the above three datasets. The quantity of GCN layers of the PLGCN was set as 1, 2, and 3, respectively. Fig. 15 describes the classification results of HPCAN applying PLGCN with different number of GCN layers. The experimental results show the best sampling efforts for the classification accuracy of HPCAN is obtained by using a two-layer GCN. The classification accuracy of the PLGCN with two GCN layers are 0.42% and 0.58% higher than that of using one GCN layer and using three GCN layers, respectively, on the IP dataset, 0.46% and 0.89% on the SA dataset, 0.95% and 1.49% on the PU dataset. When applied HPCAN using the PLGCN with too many GCN layers, the acquired features will be oversmoothing so that their own characteristics are lost and become difficult to distinguish. When applied HPCAN using the PLGCN with one GCN layer, the spatial distribution of features changes inconspicuously, and the edges between classes are still not

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:16:59 UTC from IEEE Xplore. Restrictions apply.

TABLE XI
CLASSIFICATION RESULTS OF APPLYING HPCAN FOR DIFFERENT AL METHODS ON IP DATASET

| Class | Random Sampling | Entropy [42] | CoreSet [44] | BADGE [52] | UncertainGCN [53] | CoreGCN [53] | PLGCN |
|---|---|---|---|---|---|---|---|
| 1 | 95.65 | 95.65 | **100.00** | 95.65 | 95.65 | 91.30 | **100.00** |
| 2 | 84.88 | **89.73** | 84.74 | 70.47 | 85.16 | 87.02 | 74.75 |
| 3 | 91.42 | 78.43 | 85.54 | **92.65** | 78.43 | 91.18 | 87.50 |
| 4 | **100.00** | 62.61 | 97.39 | 97.39 | 93.04 | **100.00** | 95.65 |
| 5 | 94.96 | **99.58** | 99.16 | 92.44 | 94.96 | 91.18 | 95.38 |
| 6 | **99.73** | 96.25 | 97.59 | 93.57 | 95.17 | 93.83 | 98.66 |
| 7 | 87.50 | **100.00** | 87.50 | **100.00** | **100.00** | 87.50 | **100.00** |
| 8 | 97.41 | 98.71 | **100.00** | 97.84 | 93.97 | **100.00** | **100.00** |
| 9 | **100.00** | **100.00** | **100.00** | **100.00** | **100.00** | **100.00** | **100.00** |
| 10 | 87.17 | 62.83 | **95.87** | 89.78 | 71.74 | 88.91 | 84.35 |
| 11 | 78.74 | 91.67 | 82.70 | 88.20 | 90.06 | 88.92 | **94.42** |
| 12 | 82.83 | 60.27 | 89.23 | 85.52 | 81.14 | **91.58** | 88.22 |
| 13 | 93.14 | **100.00** | **100.00** | 82.35 | 98.04 | 93.14 | 98.04 |
| 14 | 98.78 | 98.18 | **99.09** | 94.22 | 97.72 | 90.58 | 96.81 |
| 15 | 82.26 | 90.86 | 82.26 | 78.49 | 60.62 | 73.66 | **96.77** |
| 16 | 97.62 | **100.00** | 97.62 | **100.00** | 97.62 | 97.62 | 97.62 |
| OA | 88.52 | 87.35 | 90.44 | 87.81 | 87.30 | 90.05 | **91.05** |
| AA | 92.01 | 89.05 | 93.67 | 91.16 | 89.56 | 91.65 | **94.26** |
| Kappa | 86.95 | 85.46 | 89.14 | 86.07 | 85.45 | 88.64 | **89.77** |

TABLE XII
CLASSIFICATION RESULTS OF APPLYING HPCAN FOR DIFFERENT AL METHODS ON SA DATASET

| Class | Random Sampling | Entropy | CoreSet | BADGE | UncertainGCN | CoreGCN | PLGCN |
|---|---|---|---|---|---|---|---|
| 1 | 99.40 | 85.79 | **100.00** | **100.00** | 98.71 | **100.00** | **100.00** |
| 2 | 99.95 | **100.00** | **100.00** | **100.00** | 98.77 | **100.00** | 99.41 |
| 3 | 99.51 | **100.00** | **100.00** | **100.00** | **100.00** | **100.00** | **100.00** |
| 4 | **100.00** | 99.03 | **100.00** | 98.48 | 98.48 | 98.76 | 95.85 |
| 5 | 97.00 | 97.75 | 96.40 | **100.00** | 85.99 | 99.55 | 90.19 |
| 6 | 99.30 | 98.15 | 99.75 | 99.90 | 99.70 | 95.40 | **100.00** |
| 7 | **100.00** | 99.78 | **100.00** | 98.46 | 99.94 | 99.67 | 99.45 |
| 8 | 89.84 | 91.09 | 92.85 | 92.56 | 96.57 | 96.32 | **97.56** |
| 9 | 99.87 | **100.00** | **100.00** | **100.00** | 96.36 | **100.00** | 99.28 |
| 10 | 99.21 | 99.33 | **100.00** | 99.94 | 94.32 | 97.99 | 97.19 |
| 11 | **100.00** | **100.00** | **100.00** | 99.80 | **100.00** | **100.00** | **100.00** |
| 12 | 97.11 | 93.31 | **99.30** | 97.21 | 86.03 | 98.70 | 95.91 |
| 13 | 98.74 | 96.44 | **100.00** | 98.74 | 92.45 | 93.34 | **100.00** |
| 14 | 83.33 | 95.54 | **100.00** | 96.90 | 92.64 | 96.51 | **100.00** |
| 15 | 96.70 | 91.08 | 91.25 | 92.78 | 93.43 | 96.39 | **97.26** |
| 16 | 98.77 | **100.00** | **100.00** | 97.21 | 99.00 | 99.89 | **100.00** |
| OA | 96.65 | 95.72 | 97.14 | 97.15 | 95.95 | 98.00 | **98.05** |
| AA | 97.42 | 96.71 | **98.72** | 98.42 | 95.77 | 98.34 | 98.26 |
| Kappa | 96.27 | 95.23 | 96.82 | 96.84 | 95.49 | 97.77 | **97.83** |

TABLE XIII
CLASSIFICATION RESULTS OF APPLYING HPCAN FOR DIFFERENT AL METHODS ON PU DATASET

| Class | Random Sampling | Entropy | CoreSet | BADGE | UncertainGCN | CoreGCN | PLGCN |
|---|---|---|---|---|---|---|---|
| 1 | 94.00 | 83.11 | 83.20 | 92.14 | 90.04 | 84.91 | **94.03** |
| 2 | 92.94 | 85.83 | 98.07 | 98.25 | 97.98 | **100.00** | 99.45 |
| 3 | 78.73 | 82.91 | **90.22** | 77.11 | 66.76 | 86.32 | 80.34 |
| 4 | 91.17 | **92.03** | 91.24 | 56.52 | 75.36 | 71.87 | 79.31 |
| 5 | **100.00** | 94.18 | **100.00** | **100.00** | **100.00** | **100.00** | **100.00** |
| 6 | 98.35 | **98.72** | 94.46 | 96.51 | 73.11 | 97.51 | 97.83 |
| 7 | 98.92 | 99.54 | 99.23 | 97.83 | 99.69 | **100.00** | 99.38 |
| 8 | 81.21 | 72.98 | 74.50 | **89.50** | 84.79 | 84.30 | 86.30 |
| 9 | 65.67 | 82.09 | **87.63** | 84.86 | 66.74 | 72.28 | 74.63 |
| OA | 91.70 | 86.69 | 92.29 | 92.08 | 88.98 | 92.72 | **94.38** |
| AA | 89.00 | 87.93 | **90.95** | 88.08 | 83.83 | 88.58 | 90.14 |
| Kappa | 89.12 | 82.91 | 89.80 | 89.46 | 85.28 | 90.28 | **92.51** |

distinct enough. Therefore, the sampling effect of the proposed method using two GCN layers reached the best, which can optimize the spatial distribution of features effectively for better sampling performance and improving the classification efforts of HSIC models.

### *B. Influence of Different Learning Rate of GCN Training*

The learning rate plays an important role in deep learning, which is a key hyperparameter. We compared the influence of different learning rates for training GCN of the PLGCN on the classification effect of HPCAN on the IP dataset, SA dataset, and PU dataset. The learning rate in our experiment was set as 0.01, 0.001, and 0.0001. Fig. 16 shows the experimental results of training GCN with different learning rates. As shown in the experimental results, the optimal classification performance is gained by training GCN with a learning rate 0.001. A higher learning rate is not conducive to the training of GCN in periods with a small number of samples, thus impacting the sampling consequence of the AL method and the final classification

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:16:59 UTC from IEEE Xplore. Restrictions apply.

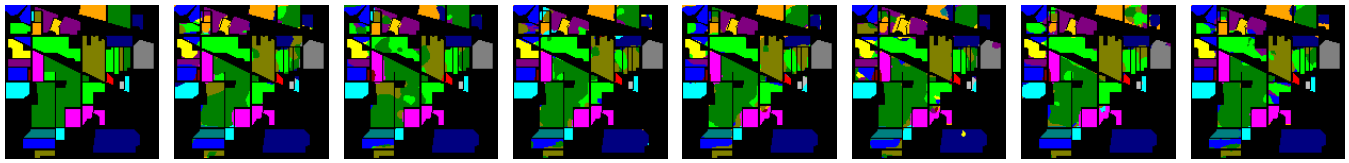



Fig. 12. Classification maps of applying HPCAN model for different AL methods on IP dataset. (a) Ground-truth, (b) random sampling, (c) entropy, (d) CoreSet, (e) BADGE, (f) UncertainGCN, (g) CoreGCN, and (h) PLGCN.

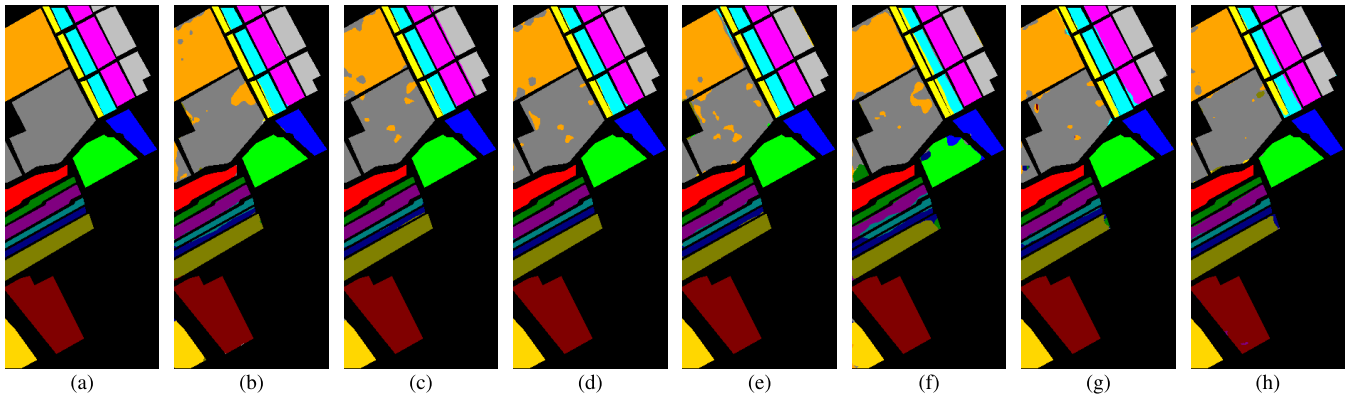



Fig. 13. Classification maps of applying HPCAN model for different AL methods on SA dataset. (a) Ground-truth, (b) random sampling, (c) entropy, (d) CoreSet, (e) BADGE, (f) UncertainGCN, (g) CoreGCN, and (h) PLGCN.

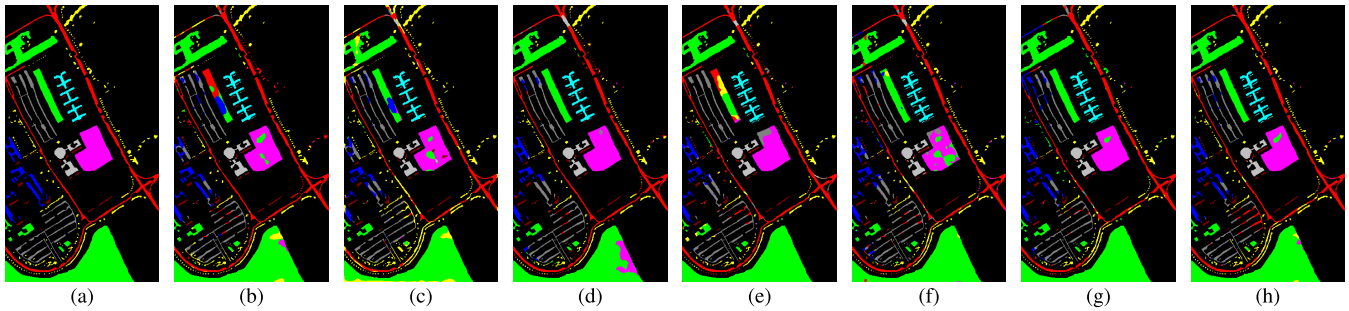



Fig. 14. Classification maps of applying HPCAN model for different AL methods on PU dataset. (a) Ground-truth, (b) random sampling, (c) entropy, (d) CoreSet, (e) BADGE, (f) UncertainGCN, (g) CoreGCN, and (h) PLGCN.

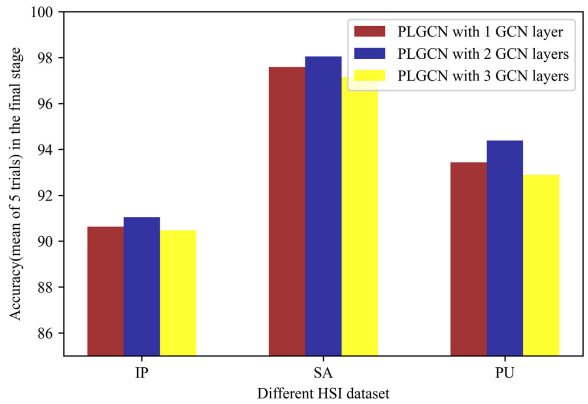



Fig. 15. Classification results of HPCAN applying PLGCN with different number of GCN layers.

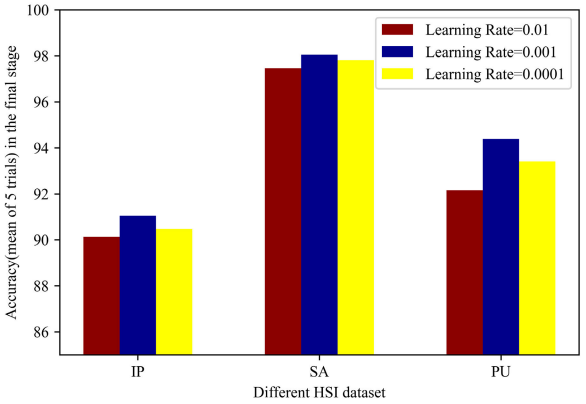



Fig. 16. Classification results of different learning rate for training GCN.

effect. The results of the learning rate set to 0.001 and 0.0001 are similar (especially on the SA dataset), but the smaller learning rate causes classification model converging with longer time. Therefore, the learning rate of 0.001 is more suitable for the training process of GCN and achieves a better sampling consequence for improving the classification results of HSIC models.

### C. Influence of the Number of GCN Training Epochs

In this section, we mensurate the effect of training GCN with different numbers of training epochs on classification performance on three datasets. The corresponding experimental results are shown in Fig. 17. The results on the SA dataset show that when the number of GCN training epochs reaches 70, the classification model HPCAN has the best

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:16:59 UTC from IEEE Xplore. Restrictions apply.

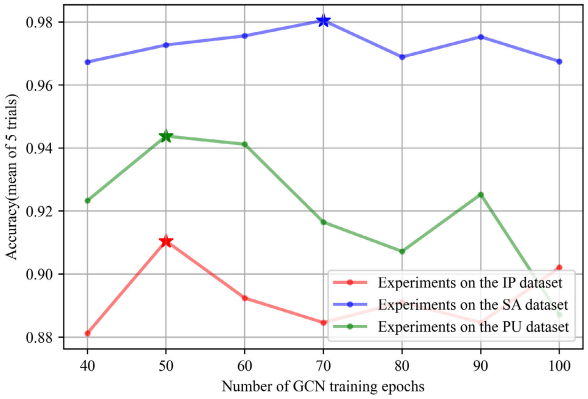



Fig. 17. Classification accuracy of HPCAN applying PLGCN with different numbers of GCN training epochs.

classification accuracy with sampling core samples adopting PLGCN. On other datasets, the best classification results are obtained when the number of GCN training epochs is 50. Due to more samples, the best number of training epochs of GCN on the SA dataset is significantly greater than that on the other two datasets. As the number of GCN training epochs increases, the features obtained by GCN are prone to the problem of oversmoothing, and the local expression ability of GCN is weakened. Therefore, with the increase in the number of GCN training epochs, the accuracy of the classification model using PLGCN shows a downward trend.

## V. Conclusion

In this article, a new AL method (called PLGCN) includes PL and GCN is proposed to solve the few shot HSI classification task, which can select more valuable and effective unlabeled samples for the training process of HSI classification models, enhance the classification accuracy of HSI classification models, and easily adapt to different HSI classification models. The prototype of each category is iteratively updated to ensure the optimality of prototype in each sampling stage; the spatial distribution of unlabeled samples is extracted via graph convolution neural network in order to obtain the better features in new space for easier discriminating. In addition, samples that are the most similar and discrepant from the prototype are selected in the improved unlabeled sample features to artificially annotate and extend a finite set of labeled samples. To prove the effectiveness of the PLGCN for improving the classification results of HSIC models, we conducted experiments compared with some canonical AL methods on three common and open-source HSI datasets. Experimental results show that samples can be obtained more balanced for each class through the PLGCN, and improve the classification performance of HSI classification models. In future research, we will focus on the improvement of GCN and use the lighter calculation to achieve the same sampling effect, to further improve the classification performance of the classification models.

## References

[1] X. Zuo, X. Yu, B. Liu, P. Zhang, and X. Tan, "FSL-EGNN: Edge-labeling graph neural network for hyperspectral image few-shot classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3165025.

[2] D. Landgrebe, "Hyperspectral image data analysis," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 17–28, Jan. 2002.

[3] S. Liu, Q. Shi, and L. Zhang, "Few-shot hyperspectral image classification with unknown classes using multitask deep learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5085–5102, Jun. 2021.

[4] J. Bai et al., "Class incremental learning with few-shots based on linear programming for hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 52, no. 6, pp. 5474–5485, Jun. 2022.

[5] X. Shen, H. Yu, C. Yu, Y. Wang, and M. Song, "Global spatial and local spectral similarity based sample augment and extended subspace projection for hyperspectral image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 3637–3640.

[6] L. Liang et al., "Estimation of crop LAI using hyperspectral vegetation indices and a hybrid inversion method," *Remote Sens. Environ.*, vol. 165, pp. 123–134, Aug. 2015.

[7] F. van der Meer, "Analysis of spectral absorption features in hyperspectral imagery," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 5, no. 1, pp. 55–68, Feb. 2004.

[8] J. Sun, X. Shen, and Q. Sun, "Hyperspectral image few-shot classification network based on the Earth mover's distance," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5534114.

[9] Z. Li, M. Liu, Y. Chen, Y. Xu, W. Li, and Q. Du, "Deep cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501618.

[10] Z. Feng, S. Yang, M. Wang, and L. Jiao, "Learning dual geometric low-rank structure for semisupervised hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 51, no. 1, pp. 346–358, Jan. 2021.

[11] P. Ghamisi et al., "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[12] J. Wang and C.-I. Chang, "Independent component analysis-based dimensionality reduction with applications in hyperspectral image analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 6, pp. 1586–1600, Jun. 2006.

[13] Q. Hou, Y. Wang, L. Jing, and H. Chen, "Linear discriminant analysis based on kernel-based possibilistic C-means for hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 8, pp. 1259–1263, Aug. 2019.

[14] J. Peng, Y. Zhou, and C. L. P. Chen, "Region-kernel-based support vector machines for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 9, pp. 4810–4824, Sep. 2015.

[15] Y. Chen, Z. Lin, and X. Zhao, "Riemannian manifold learning based k-nearest-neighbor for hyperspectral image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2013, pp. 1975–1978.

[16] Y. Bazi and F. Melgani, "Gaussian process approach to remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 1, pp. 186–197, Jan. 2010.

[17] Y. Zhao, L. Ha, H. Wang, and X. Ma, "Few-shot class incremental learning for hyperspectral image classification based on constantly updated classifier," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 1376–1379.

[18] G. Cheng, Z. Li, J. Han, X. Yao, and L. Guo, "Exploring hierarchical convolutional features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 11, pp. 6712–6722, Nov. 2018.

[19] L. He, J. Li, C. Liu, and S. Li, "Recent advances on spectral–spatial hyperspectral image classification: An overview and new guidelines," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1579–1597, Mar. 2018.

[20] J. Li, P. R. Marpu, A. Plaza, J. M. Bioucas-Dias, and J. A. Benediktsson, "Generalized composite kernel framework for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 9, pp. 4816–4829, Sep. 2013.

[21] J. Huang, K. Liu, M. Xu, M. Perc, and X. Li, "Background purification framework with extended morphological attribute profile for hyperspectral anomaly detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 8113–8124, 2021.

[22] W. Li, C. Chen, H. Su, and Q. Du, "Local binary patterns and extreme learning machine for hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3681–3693, Jul. 2015.

[23] Y. Zhang, K. Liu, Y. Dong, K. Wu, and X. Hu, "Semisupervised classification based on SLIC segmentation for hyperspectral image," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 8, pp. 1440–1444, Aug. 2020.

[24] J. Bai et al., "Few-shot hyperspectral image classification based on adaptive subspaces and feature transformation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5523917.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:16:59 UTC from IEEE Xplore. Restrictions apply.

[25] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[26] Y. Chen, X. Zhao, and X. Jia, "Spectral–spatial classification of hyperspectral data based on deep belief network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2381–2392, Jun. 2015.

[27] L. Mou, P. Ghamisi, and X. X. Zhu, "Deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3639–3655, Jul. 2017.

[28] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[29] J. Yue, W. Zhao, S. Mao, and H. Liu, "Spectral–spatial classification of hyperspectral images using deep convolutional neural networks," *Remote Sens. Lett.*, vol. 6, no. 6, pp. 468–477, Jun. 2015.

[30] B. Liu, X. Yu, A. Yu, and G. Wan, "Deep convolutional recurrent neural network with transfer learning for hyperspectral image classification," *J. Appl. Remote Sens.*, vol. 12, no. 2, p. 1, Jun. 2018.

[31] Y. Li, H. Zhang, and Q. Shen, "Spectral–spatial classification of hyperspectral imagery with 3D convolutional neural network," *Remote Sens.*, vol. 9, no. 1, p. 67, Jan. 2017.

[32] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[33] M. E. Paoletti et al., "Capsule networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2145–2160, Apr. 2019.

[34] J. Wang, S. Guo, R. Huang, L. Li, X. Zhang, and L. Jiao, "Dual-channel capsule generation adversarial network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501016.

[35] L. Zhu, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Generative adversarial networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5046–5063, Sep. 2018.

[36] X. He and Y. Chen, "Optimized input for CNN-based hyperspectral image classification using spatial transformer network," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 12, pp. 1884–1888, Dec. 2019.

[37] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[38] H. Sun, X. Zheng, X. Lu, and S. Wu, "Spectral–spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3232–3245, 2020.

[39] M. Zhu, L. Jiao, F. Liu, S. Yang, and J. Wang, "Residual spectral–spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 449–462, Jan. 2021.

[40] C. Ding et al., "Integrating hybrid pyramid feature fusion and coordinate attention for effective small sample hyperspectral image classification," *Remote Sens.*, vol. 14, no. 10, p. 2355, May 2022. [Online]. Available: https://www.mdpi.com/2072-4292/14/10/2355

[41] T. Song, Y. Wang, C. Gao, H. Chen, and J. Li, "MSLAN: A two-branch multidirectional spectral–spatial LSTM attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5528814.

[42] D. Wang and Y. Shang, "A new active labeling method for deep learning," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jul. 2014, pp. 112–119.

[43] Y. Gal, R. Islam, and Z. Ghahramani, "Deep Bayesian active learning with image data," in *Proc. 34th Int. Conf. Mach. Learn.*, vol. 70, 2017, pp. 1183–1192.

[44] O. Sener and S. Savarese, "Active learning for convolutional neural networks: A core-set approach," in *Proc. Int. Conf. Learn. Represent.*, 2018. [Online]. Available: https://openreview.net/forum?id=H1aIuk-RW

[45] J. M. Haut, M. E. Paoletti, J. Plaza, J. Li, and A. Plaza, "Active learning with convolutional neural networks for hyperspectral image classification using a new Bayesian approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 11, pp. 6440–6461, Nov. 2018.

[46] D. Yoo and I. S. Kweon, "Learning loss for active learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 93–102.

[47] R. Sheikh, A. Milioto, P. Lottes, C. Stachniss, M. Bennewitz, and T. Schultz, "Gradient and log-based active learning for semantic segmentation of crop and weed for agricultural robots," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2020, pp. 1350–1356.

[48] D. Gudovskiy, A. Hodgkinson, T. Yamaguchi, and S. Tsukizawa, "Deep active learning for biased datasets via Fisher kernel self-supervision," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9038–9046.

[49] S. Sinha, S. Ebrahimi, and T. Darrell, "Variational adversarial active learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 5971–5980.

[50] B. Zhang, L. Li, S. Yang, S. Wang, Z.-J. Zha, and Q. Huang, "State-relabeling adversarial active learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 8753–8762.

[51] Y. Ren, B. Wang, J. Zhang, and Y. Chang, "Adversarial active learning based heterogeneous graph neural network for fake news detection," in *Proc. IEEE Int. Conf. Data Mining (ICDM)*, Nov. 2020, pp. 452–461.

[52] J. T. Ash, C. Zhang, A. Krishnamurthy, J. Langford, and A. Agarwal, "Deep batch active learning by diverse, uncertain gradient lower bounds," in *Proc. Int. Conf. Learn. Represent.*, 2020. [Online]. Available: https://openreview.net/forum?id=ryghZJBKPS

[53] R. Caramalau, B. Bhattarai, and T.-K. Kim, "Sequential graph convolutional network for active learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 9578–9587.

[54] J. Li, "Active learning for hyperspectral image classification with a stacked autoencoders based neural network," in *Proc. 7th Workshop Hyperspectral Image Signal Process., Evol. Remote Sens. (WHISPERS)*, Jun. 2015, pp. 1–4.

[55] Z. Sun, M. Chi, and J. A. Benediktsson, "Computational efficiency active learning for classification of hyperspectral images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 5138–5140.

[56] J. Wang and Y. Zhai, "Prototypical Siamese networks for few-shot learning," in *Proc. IEEE 10th Int. Conf. Electron. Inf. Emergency Commun. (ICEIEC)*. Red Hook, NY, USA: Curran Associates Inc., Jul. 2020, pp. 178–181.

[57] C. Zhuang and Q. Ma, "Dual graph convolutional networks for graph-based semi-supervised classification," in *Proc. World Wide Web Conf. World Wide Web (WWW)*, 2018, pp. 499–508, doi: 10.1145/3178876.3186116.

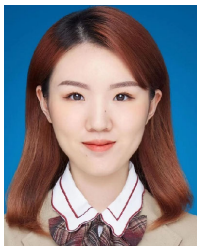
[58] V. Nair and G. E. Hinton, "Rectified linear units improve restricted Boltzmann machines," in *Proc. 27th Int. Conf. Int. Conf. Mach. Learn.*, Madison, WI, USA: Omnipress, 2010, pp. 807–814.

[59] S. K. Roy, S. Manna, T. Song, and L. Bruzzone, "Attention-based adaptive spectral–spatial kernel ResNet for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7831–7843, Sep. 2021.



**Chen Ding** (Member, IEEE) received the Ph.D. degree in computer science and technology from Northwestern Polytechnical University, Xi'an, China, in 2018.

He is an Associate Professor with the School of Computer Science, Xi'an University of Posts and Telecommunications, Xi'an. Approved in 2019, he presided over the Youth Program of the National Natural Science Foundation of China. His research interests include image processing, machine learning, and pattern recognition.



**Mengmeng Zheng** received the bachelor's degree from the Xi'an University of Posts and Telecommunications, Xi'an, China, in 2020, where she is currently pursuing the M.S. degree, under the supervision of Dr. C. Ding.

Her research interests include hyperspectral image processing and deep learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:16:59 UTC from IEEE Xplore. Restrictions apply.



**Sirui Zheng** is currently pursuing the bachelor's degree with the Xi'an University of Posts and Telecommunications, Xi'an, China, under the guidance of Dr. C. Ding.

His research interests include hyperspectral image processing and deep learning.



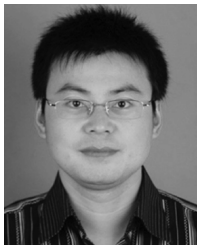
**Yaoyang Xu** is currently pursuing the bachelor's degree with the Xi'an University of Posts and Telecommunications, Xi'an, China, under the guidance of Dr. C. Ding.

His research interests include hyperspectral image processing and deep learning.



**Lei Zhang** (Member, IEEE) received the Ph.D. degree in computer science and technology from Northwestern Polytechnical University, Xi'an, China, in 2018.

He was a Research Staff with the School of Computer Science, The University of Adelaide, Adelaide, SA, Australia, from 2017 to 2019. He was a Research Scientist with the Inception Institute of Artificial Intelligence (IIAI), Abu Dhabi, United Arab Emirates, from 2019 to 2020. He is also a Professor with the School of Computer Science, Northwestern Polytechnical University. His research interests include image processing, machine learning, and video analysis.



**Wei Wei** (Senior Member, IEEE) received the Ph.D. degree from Northwestern Polytechnical University, Xi'an, China, in 2012.

He is currently a Professor with the School of Computer Science and Engineering, Northwestern Polytechnical University. He has authored more than 40 articles, including IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, *International Journal of Computer Vision*, and IEEE TRANSACTIONS ON IMAGE PROCESSING. His research interests include image processing, machine learning, and pattern recognition.

Dr. Wei is an Associate Editor of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.



**Yanning Zhang** (Senior Member, IEEE) received the B.S. degree from the Dalian University of Technology, Dalian, China, in 1988, the M.S. degree from the School of Electronic Engineering, Northwestern Polytechnical University, Xi'an, China, in 1993, and the Ph.D. degree from the School of Marine Engineering, Northwestern Polytechnical University, in 1996.

She is a Professor with the School of Computer Science, Northwestern Polytechnical University. She is also a Cheung Kong Professor of the Ministry of Education, Beijing, China. She has authored more than 200 articles. Her research interests include remote sensing image analysis, computer vision, and pattern recognition.

Dr. Zhang is an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and *Pattern Recognition*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:16:59 UTC from IEEE Xplore. Restrictions apply.